

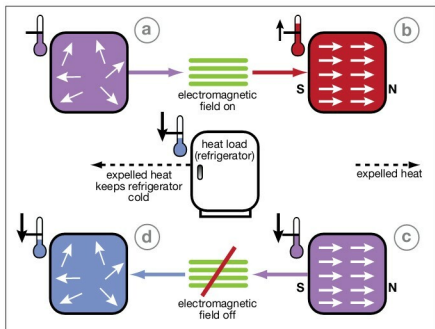


Fig. 2: Magnetic refrigeration cycle. A magnetocaloric material heats up when magnetised (b); if cooled and then demagnetised (c), its temperature drops dramatically (d) (Courtesy: Talbott, NIST)

the excess disorder, and this becomes heat. Conversely, when the magnetic field is again removed, the substance becomes cold. The heat is transferred to a fluid that is pumped back and forth past the substance inside a cylinder. The end that becomes cold will be located inside the refrigerator and the warm end will be outside.

The first milestone in magnetic cooling has been achieved between 5°C and 10°C of cooling and the figure is currently at 8.7°C. This means that a refrigerator at room temperature (20°C) can be cooled to almost 11°C. Of course, this is not quite enough, but the objective of conducting research is to test different materials, varying operating conditions and the strength of the magnetic field. Despite the technology being potentially energy-efficient and environment-friendly, it is probably not realistic to think that magnetic cooling technology will be used in homes right away.

Thermoelectric refrigeration: Cooling in miniature

In conventional cooling devices, heat is carried away by a working fluid

like a CFC, which involves the moving parts that cause most equipment breakdowns, environmental damage and bulkiness. In thermoelectric devices, the working fluid is electrical current that runs through a junction between differently doped semiconductors and pulls heat away from that junction, producing cooling without any moving part.

Current thermoelectric materials operate at roughly 10 per cent of Carnot efficiency—the theoretical maximum allowed by the laws of thermodynamics—compared to about 30 per cent for an average household refrigerator. The theory behind thermoelectric devices has been around for more than 40 years, but current materials do not rival the efficiency of compressor-based devices.

Such thermoelectric devices already exist in consumer products like plug-in auto beverage coolers, where energy efficiency is less important than portability and low weight. Researchers are facing the challenge of finding new materials that could take the technology to the next level at which the efficiency would rival that of conventional coolants in air-condi-

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